

Christopher Musco

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Education

Massachusetts Institute of Technology

Ph.D. Candidate, Computer Science – Theory of Computation

Advisor: Jonathan Kelner

Masters Thesis: Dimensionality Reduction for Sparse and Structured Matrices (Spring 2015)

Cambridge, MA

2013 – present

Yale University

B.S. Applied Mathematics, B.S. Computer Science

Applied Mathematics Thesis: Fast Discrete Laplace Transforms (with Vladimir Rokhlin)

Computer Science Thesis: Graph Constructions for Machine Learning (with Daniel Spielman)

New Haven, CT

2008 – 2012

Research Areas

theory of algorithms • scalable machine learning • randomized linear algebra • low rank approximation
dimensionality reduction • sketching and streaming • algorithmic graph theory • iterative matrix algorithms

Publications

Authors appear in alphabetical order, in the tradition of mathematics and theoretical computer science.

Conference and Journal Papers.....

Random Fourier Features for Kernel Ridge Regression. Haim Avron, Michael Kapralov, Cameron Musco, Christopher Musco, Ameya Velingker, Amir Zandieh. *ICML 2017*.

Input Sparsity Time Low-Rank Approximation via Ridge Leverage Score Sampling. Michael B. Cohen, Cameron Musco, Christopher Musco. *SODA 2017*.

Determining Tournament Payout Structures for Daily Fantasy Sports. Christopher Musco, Maxim Sviridenko, Justin Thaler. *ALENEX 2017*.

Principal Component Projection Without Principal Component Analysis. Roy Frostig, Cameron Musco, Christopher Musco, Aaron Sidford. *ICML 2016*.

Randomized Block Krylov Methods for Stronger and Faster Approximate Singular Value Decomposition. Cameron Musco, Christopher Musco. *NIPS 2015*. Invited for full oral presentation (1 of 15 out of 403 accepted papers). Also presented at the 2016 Copper Mountain Conference on Iterative Methods

Dimensionality Reduction for K-Means Clustering and Low Rank Approximation. Michael B. Cohen, Samuel Elder, Cameron Musco, Christopher Musco, Madalina Persu. *STOC 2015*.

Principled Sampling for Anomaly Detection. Brendan Juba, Fan Long, Christopher Musco, Stelios Sidiropoulos-Douskos, Martin Rinard. *NDSS 2015*.

Uniform Sampling for Matrix Approximation. Michael B. Cohen, Yin Tat Lee, Cameron Musco, Christopher Musco, Richard Peng, Aaron Sidford. *ITCS 2015*.

Single Pass Spectral Sparsification in Dynamic Streams. Michael Kapralov, Yin Tat Lee, Cameron Musco, Christopher Musco, Aaron Sidford. *FOCS 2014, SIAM Journal on Computing 2017*.

Other.....

Recursive Sampling for the Nyström Method. Cameron Musco, Christopher Musco. *In submission*.

Stability of the Lanczos Method for Matrix Function Approximation. Cameron Musco, Christopher Musco, Aaron Sidford. *In submission.*

Teaching and Mentorship

Advanced Algorithms (MIT 6.854/18.415)

Teaching Assistant

2016

Held weekly office hours, wrote problem sets, and compiled notes and resources for an updated version of this capstone graduate algorithms course. Wrote guide on topics and expectations for student projects. Prepared and delivered a lecture on linear programming relaxations.

Technical Communication Skills for Graduate Students (MIT 6.S977)

Workshop Leader

2016

Lead weekly workshops that supplemented lecture material with hands-on writing and presentation exercises. Met with students one-on-one outside of class to offer individual support and coaching in technical communication.

MIT Graduate Communication Lab

Communication Advisor

2015 – present

Helping to establish an EECS Communication Lab to provide over 700 graduate students with support in academic writing, oral presentation, job applications, and generally, in communicating effectively. Offer feedback, guidance, and coaching to individual students during weekly office hours (75+ individual appointments to date).

CSAIL Algorithms Office Hours

Student Member

2016 – present

Meet with researchers, generally in applied fields, who seek advice on solving or framing algorithmic problems that appear in their research.

Research Science Institute at MIT

Summer Research Advisor

2014

Mentored Christopher Wang, a high school student participating in the prestigious RSI summer program. Chris' project, "Relaxation of a Concurrent Disjoint-Set", was selected as one of 10 program finalists.

Introduction to Computer Science (Yale CPSC-201)

Teaching Assistant

2012

Covered functional programming, boolean algebra, formal language theory, and basic complexity theory.

Introduction to Programming (Yale CPSC-112)

Teaching Assistant

2011

Led weekly section for 15-25 students covering basic Java programming. Tutored 6 students individually.

Introductory Calculus (Yale MATH-112 & MATH-115)

Grader

2009 – 2010

Scored problem sets and provided explanations and solutions for students, as well as feedback to the professor.

Industry Research

Yahoo Labs

New York, NY

Research Intern, Scalable Machine Learning Group

Summer 2015

Mentors: Justin Thaler, Maxim Srividenko, Edo Liberty

Worked on theoretical matrix sampling problems and on algorithms for managing Yahoo's Fantasy Sports platform.

Talks

Sketching and Sampling Methods for Large Scale Linear Algebra

Twitter, Cambridge

July 2017

Recursive Sampling for the Nyström Method	
New England Machine Learning Day Poster Session	May 2017
Algorithms for Determining Tournament Payout Structures	
Meeting on Algorithm Engineering and Experiments (ALENEX)	Jan. 2017
Introduction to Linear Sketching	
MIT, Theory Retreat	Sept. 2016
Principal Component Projection without Principal Component Analysis	
International Conference on Machine Learning (ICML)	June 2016
Ridge Leverage Score Sampling	
National Institute of Informatics, Shonan Meeting	July 2016
MIT, Algorithms & Complexity Seminar	July 2016
University of Utah, Data Group Meeting	Jan. 2016
Randomized Block Krylov Methods	
Neural Information Processing Systems (NIPS) Poster Session	Dec. 2015
Large Scale Column Subset Selection	
Yahoo Labs, Science Week Poster Session	July 2015
Dimensionality Reduction for K-Means Clustering	
IBM T.J. Watson, Mathematical Sciences Research Seminar	Aug. 2015
Symposium on Theory of Computing (STOC)	June 2015
MIT, Theory Lunch	June 2014
Principled Sampling for Anomaly Detection	
MIT, Defense Advanced Research Projects Agency Site Visit	Mar. 2015
Network and Distributed System Security Symposium (NDSS)	Feb. 2015
Uniform Sampling for Matrix Approximation	
Yahoo Labs, Scalable Machine Learning Research Seminar	June 2015
Single Pass Spectral Sparsification in Dynamic Streams	
MIT, Annual Sublinear Algorithms Day Poster Session	Apr. 2015
Harvard University, Theory Seminar	Nov. 2014
Foundations of Computer Science (FOCS)	Oct. 2014

Service

External Reviewer.....	
International Workshop on Randomization and Computation (RANDOM)	2017
SIAM Journal on Matrix Analysis and Applications (SIMAX)	2016, 2017
IEEE Transactions on Knowledge and Data Engineering (TKDE)	2017
Conference on Learning Theory (COLT)	2016
Symposium on Discrete Algorithms (SODA)	2016, 2017, 2018
IEEE Transactions on Signal Processing (TSP)	2016, 2017, 2018
Neural Information Processing Systems (NIPS)	2015, 2016, 2017
European Symposia on Algorithms (ESA)	2015, 2017
Foundations of Computer Science (FOCS)	2015

International Colloquium on Automata, Languages and Programming (ICALP)	2015, 2017
Network and Distributed System Security Symposium (NDSS)	2015

Other Research Experience

Japan National Institute of Informatics	2016
<i>Invited participant, Shonan meeting on Recent Advances in Randomized Numerical Linear Algebra</i>	

Honors and Awards

National Science Foundation	2014-2017
<i>Graduate Research Fellowship, recipient.</i>	
Yale University	2012
<i>Cum Laude, Distinction in Both Majors.</i>	

Professional Experience

Redfin	Seattle, WA
<i>Software Engineer, Data Team</i>	2012 – 2014
Developed backend Java infrastructure for internet-powered real estate startup. Incorporated large-scale machine learning algorithms into production code. Managed three interns and onboarded four new-hires.	

Elysium Digital	Cambridge, MA
<i>Summer Consultant</i>	Summer 2011
Provided technical expertise in software patent cases at leading litigation consulting company.	

Amicus	New Haven, CT
<i>Software Developer</i>	2010 – 2011
Built tools for political campaign management and social fundraising at an early-stage startup that went on to raise nearly 4 million dollars in seed funding.	

Moody's Investors Service	New York, NY
<i>Intern, Asset Finance Group</i>	Summer 2009
Established and monitored bond ratings on securitizations in three asset classes. Wrote code to automate analysis for student loan data and to compile records on aircraft leases.	

U.S. Department of Agriculture	East Greenwich, RI
<i>Intern, Natural Resources Conservation Service</i>	Summer 2007
Collected and analyzed data for two major state conservation and research projects. Integrated mapping software and database tools to develop a system for compiling and presenting soil data.	